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Academic year 2024-2025 (Odd Sem)

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

Date	13th Jan 2026	Maximum Marks	(10+50) Marks
Course Code	IS353IA	Duration	(30+90) = 120 Min
Sem	V Sem	Improvement CIE	
UG/PG	UG	Faculty: AS/VH/VG/SB/SGR/MEM	

Artificial Intelligence and Machine Learning
(Common to CSE/ISE/CD/CY)

Q. No.	QUIZ	M	BT	CO
1.	What is the primary drawback of a standard Hill-Climbing Search algorithm, and what is the simplest modification used to mitigate it?	2	L1	CO1
2.	Briefly explain the roles of the two main operators, Crossover and Mutation, in a Genetic Algorithm's generation of a new population.	2	L2	CO1
3.	In the context of Unsupervised Cluster Evaluation, define the two concepts of Cohesion and Separation.	2	L1	CO2
4.	Distinguish between Hard Clustering and Soft Clustering in terms of data point assignment. Give one algorithm example for each.	2	L3	CO2
5.	The Bisecting K-means algorithm is a top-down hierarchical approach. How does it improve upon the stability and quality of results compared to a single run of standard K-means?	2	L3	CO5

Q. No.	TEST	M	BT	CO
1.	Suppose that the data mining task is to cluster the following eight points with (x, y) representing location into three clusters: A1(2, 10), A2(2, 5), A3(8, 4), B1(5, 8), B2(7, 5), B3(6, 4), C1(1, 2), C2(4, 9): The distance function is Euclidean distance. Suppose initially we assign A1, B1, and C1 as the center of each cluster, respectively. a) Write down k-means algorithm b) Apply k-means algorithm for the three cluster centers after the first round execution	10	L4	CO1
2. a	How the distance between clusters is measured in hierarchical clustering? Explain the use of this measure in making decision on when to stop the iteration	5	L2	CO1
b	Explain the primary objectives of Cluster Evaluation and the three main types of validity measures.	5	L2	CO2
3. a	What is the main dilemma when "Choosing a Cluster Validity Measure"? Outline the practical trade-offs involved.	5	L3	CO2
b	Explain two different types of clusters based on their geometry or structure, and name a suitable clustering algorithm for each.	5	L2	CO2

4. a	The goal is to maximize the value of a function f(x) where x ∈{1, 2, 3, 4, 5} and the values are f(1)=3, f(2)=7, f(3)=5, f(4)=9, f(5)=6. Show the trace of Steepest-Ascent Hill-climbing starting at x=1. Assume neighbours are x ±1.	5	L2	CO2
b	Describe the process and main operators of a Genetic Algorithm (GA).	5	L2	CO1
5. a	Explain the core principle and mechanism of Simulated Annealing. How does it guarantee completeness (eventual convergence to the global optimum)?	5	L3	CO5
b	Define Local Search Algorithms and explain why they are essential for solving Optimization Problems in large search spaces.	5	L2	CO2

Course Outcomes: After completing the course, the students will be able to: -	
CO1:	Explain and apply AI and ML algorithms to address various requirements of real-world problems
CO2:	Design and develop AI and ML solutions to benefit society, science, and industry.
CO3:	Use modern tools to create AI and ML solutions.
CO4:	Demonstrate effective communication through team presentations and reports to analyze the impact of AI and ML solutions on society and nature.
CO5:	Conduct performance evaluation, modeling, and validation of AI and ML solutions benefiting lifelong learning

Academic year 2024-2025 (Odd Sem)

Date: 13/01/2026	Improvement CIE	Max. Marks: 60
Semester: V	UG	Duration: 2 Hours
Course: Artificial Intelligence and Machine Learning (Common to CSE/ISE/CD/CY)		Code: IS353IA

Q.No	Quiz	M
1.	**Primary Drawback:** The algorithm is guaranteed to get stuck in a **local optimum** (a state better than its neighbours, but not the global best) and cannot backtrack or explore the search space further. **Modification:** The simplest modification is to use **Stochastic Hill-Climbing**, where a move is chosen randomly from among the uphill moves, or **Random-Restart Hill-Climbing**, which runs the algorithm multiple times from different random initial states.	2
2.	**Crossover (Recombination):** This operator combines the "genetic material" (components/attributes) of two parent states to create one or more offspring states. It is the primary mechanism for combining desirable traits from good solutions, driving **large-scale changes** and **exploration** of the solution space. **Mutation:** This operator randomly alters one or more components of a newly created state (offspring). Its role is to introduce small, random **diversity** into the population, preventing the algorithm from getting stuck in local optima by ensuring that no part of the search space is entirely excluded from consideration.	2
3.	**Cohesion:** Measures how **closely related** the data points are *within* a cluster. High cohesion (small within-cluster distances) indicates a desirable, tight cluster. Metrics often use the **WCSS** (Within-Cluster Sum of Squares). **Separation:** Measures how **distinct or far apart** one cluster is from other clusters. High separation (large between-cluster distances) indicates a desirable, well-isolated cluster. Metrics often use the distance between cluster centroids.	2
4	**Hard Clustering:** Each data point is assigned **exclusively** to exactly one cluster. o *Example:* **K-means** (uses nearest centroid distance). **Soft Clustering:** Each data point is assigned a **degree of membership** or a probability of belonging to *every* cluster. o *Example:* **Gaussian Mixture Models (GMM)** or **Fuzzy C-Means**.	2

5	**Improvement:** It overcomes the initialization issue of standard K-means by performing many independent K=2 runs (bisections) on the same cluster and choosing the one that yields the lowest SSE. **Result:** This iterative, constrained approach makes the final clustering **less susceptible to poor initial centroid placement** and typically results in a **lower overall SSE** (closer to the global optimum) than a single random-start K-means run.	2

Q.No	Quiz	M

1.	**Algorithm 1.5** K-Means
	Cluster($\mathbf{X}$) {Cluster dataset $\mathbf{X}$}
	Initialize cluster centers μ_j for $j = 1, \ldots, k$ randomly
	repeat
	for $i = 1$ **to** m **do**
	Compute $j' = \text{argmin}_{j=1,\ldots,k}\, d(x_i, \mu_j)$
	Set $r_{ij'} = 1$ and $r_{ij} = 0$ for all $j' \neq j$
	end for
	for $j = 1$ **to** k **do**
	Compute $\mu_j = \frac{\sum_i r_{ij} x_i}{\sum_i r_{ij}}$
	end for
	until Cluster assignments r_{ij} are unchanged
	return $\{\mu_1, \ldots, \mu_k\}$ and r_{ij}
	
	d(a,b) denotes the Eucledian distance between a and b. It is obtained directly from the distance matrix or calculated as follows: d(a,b)=sqrt($(x_b-x_a)^2+(y_b-y_a)^2$))
	seed1=A1=(2,10), seed2=A4=(5,8), seed3=A7=(1,2)
	
	epoch1 – start:
	
	A1:
	d(A1, seed1)=0 as A1 is seed1
	d(A1, seed2)= $\sqrt{13}$ >0
	d(A1, seed3)= $\sqrt{65}$ >0
	➔A1 ∈ cluster1
	
	A2:
	d(A2,seed1)= $\sqrt{25}$ = 5
	d(A2, seed2)= $\sqrt{18}$ = 4.24
	d(A2, seed3)= $\sqrt{10}$ = 3.16 ← smaller
	➔ A2 ∈ cluster3
	
	A3:
	d(A3, seed1)= $\sqrt{36}$ = 6
	d(A3, seed2)= $\sqrt{25}$ = 5 ← smaller
	d(A3, seed3)= $\sqrt{53}$ = 7.28
	➔ A3 ∈ cluster2
	
	A4:
	d(A4, seed1)= $\sqrt{13}$
	d(A4, seed2)=0 as A4 is seed2
	d(A4, seed3)= $\sqrt{52}$ >0
	➔ A4 ∈ cluster2
	
	A5:
	d(A5, seed1)= $\sqrt{50}$ = 7.07
	
	A6:
	d(A6, seed1)= $\sqrt{52}$ = 7.21
		3
		
		5

	d(A5, seed2)= $\sqrt{13}$ = 3.60 ← smaller d(A5, seed3)= $\sqrt{45}$ = 6.70 → A5 ∈ cluster2 d(A6, seed2)= $\sqrt{17}$ = 4.12 ← smaller d(A6, seed3)= $\sqrt{29}$ = 5.38 → A6 ∈ cluster2 A7: d(A7, seed1)= $\sqrt{65}$ >0 d(A7, seed2)= $\sqrt{52}$ >0 d(A7, seed3)=0 as A7 is seed3 → A7 ∈ cluster3 end of epoch1 A8: d(A8, seed1)= $\sqrt{5}$ d(A8, seed2)= $\sqrt{2}$ ← smaller d(A8, seed3)= $\sqrt{58}$ → A8 ∈ cluster2 new clusters: 1: {A1}, 2: {A3, A4, A5, A6, A8}, 3: {A2, A7} b) centers of the new clusters: C1= (2, 10), C2= ((8+5+7+6+4)/5, (4+8+5+4+9)/5) = (6, 6), C3= ((2+1)/2, (5+2)/2) = (1.5, 3.5)	2
2 a	In hierarchical clustering, the distance between clusters is measured using **linkage methods**, which define how dissimilarity between two clusters is computed based on distances among their data points. Common linkage methods include **single linkage (minimum distance)**, which considers the closest pair of points and may produce elongated clusters; **complete linkage (maximum distance)**, which uses the farthest pair of points and tends to form compact clusters; **average linkage**, which computes the mean of all pairwise distances between clusters; **centroid linkage**, which measures the distance between cluster centroids; and **Ward's method**, which defines distance as the increase in within-cluster variance caused by merging two clusters and is effective in producing compact, homogeneous clusters. The cluster distance measure plays a key role in deciding when to stop the hierarchical clustering process, which is commonly visualized using a **dendrogram**. The vertical axis of the dendrogram represents the distance at which clusters are merged. A common stopping criterion is to identify a **large jump (gap)** in the merge distances, indicating that further merging would combine dissimilar clusters. The hierarchy is then cut at this level to obtain the final clusters. Alternatively, clustering can be stopped by setting a **predefined distance threshold**, beyond which no further cluster merging is allowed.	5
2 b	The primary objective of **Cluster Evaluation** or **Cluster Validity** is to assess the quality of the clustering results. Since clustering is an unsupervised task, there is no ground truth, making evaluation essential. Key objectives include: • **Determining the Clustering Tendency:** Assessing if the data naturally contains non-random structure or clusters, or if it's just uniform noise. • **Evaluating the Quality of a Single Clustering:** Measuring how well the points are grouped, typically based on high **cohesion** (similarity within clusters) and high **separation** (dissimilarity between clusters). • **Comparing Different Clustering Results:** Choosing the best result among multiple runs of the same algorithm (e.g., with different parameters) or across different algorithms.	5

- **Determining the Optimal Number of Clusters (k):** Identifying the value of k that provides the most meaningful partition of the data.

The three main types of cluster validity measures are:

1. **External Measures (Supervised):** Used when the ground truth (predefined class labels) is available. They compare the clustering results to the external labels.
2. **Internal Measures (Unsupervised):** Used when no external information is available. They evaluate the clustering structure using only the data itself (e.g., cohesion and separation).
3. **Relative Measures:** Used to compare two different clusterings or to find the optimal number of clusters for a specific algorithm.

3 a 5

The main dilemma in choosing a cluster validity measure stems from the inherent nature of clustering: **which criteria of 'goodness' truly align with the goal of the analysis?** Different metrics are based on different assumptions about what constitutes a good cluster (e.g., spherical, dense, hierarchical).

Trade-off Dimension	**Internal Measures**	**External Measures**
Availability of Information	Applicable always (unsupervised).	Only applicable when ground truth labels exist (supervised).
Criterion of "Goodness"	Focus on *structural* properties (Cohesion and Separation). Results are only meaningful relative to the metric's assumption (e.g., SSE favors spherical clusters).	Focus on *agreement* with a known, external standard. Results are highly objective.
Utility	Best for comparing different parameter settings of one algorithm (k).	Best for comparing different algorithms against a known benchmark.
Computational Cost	Varies. Some (like SSE) are fast, others (like Silhouette) are $O(N^2)$ for large data.	Typically fast once the contingency matrix is built ($O(N)$ or $O(k_{clusters}^2)$).

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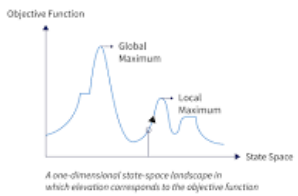
3b	Clustering algorithms make different assumptions about the shape and structure of the clusters they are trying to find. **Centroid/Prototype-Based Clusters (e.g., Spherical Clusters):** **Description:** These clusters are often **spherical** (or globular) and convex, defined by their central point, or **centroid** (the mean) or **medoid** (an actual data point). All points within the cluster are relatively close to this central point. **Characteristic:** They assume roughly equal size and density and are typically well-separated. **Suitable Algorithm: K-means**. K-means works by minimizing the squared distance between points and their assigned centroid, naturally favoring a spherical shape. **Density-Based Clusters (e.g., Arbitrarily Shaped Clusters):** **Description:** These clusters are defined as contiguous regions of high point density, separated by regions of low point density (noise or outliers). This allows for the discovery of clusters with **arbitrary shapes** (e.g., L-shaped, curved, or inter-twined) that centroid-based methods would fail to capture. **Characteristic:** They can identify clusters of different sizes and shapes, and are robust to noise and outliers (which are typically classified as low-density points). **Suitable Algorithm: DBSCAN (Density-Based Spatial Clustering of Applications with Noise)**. DBSCAN groups together points that are closely packed (within a neighborhood radius)and have a minimum number of neighbor	5

4 a	**Problem Setup**:
	
	• Objective: Maximize (f(x)) for (x \in {1, 2, 3, 4, 5}).
	• Function values: (f(1) = 3), (f(2) = 7), (f(3) = 5), (f(4) = 9), (f(5) = 6).
	• Neighbors: (x \pm 1) (within bounds; e.g., (x=1) has only neighbor 2).
	• Algorithm: From the current state, evaluate all neighbors and move to the one with the highest (f)-value if it exceeds the current value. Stop if no better neighbor exists.
	
	Trace Starting at (x=1):
	
	1. **Current: (x=1), (f(1)=3)**
	  o Neighbors: {2}.
	  o (f(2)=7 > 3).
	  o Steepest ascent: Move to (x=2).
	2. **Current: (x=2), (f(2)=7)**
	  o Neighbors: {1, 3}.
	  o (f(1)=3 < 7), (f(3)=5 < 7).
		5

	○ No better neighbor. **Stop.** ○ Local maximum found: (x=2), (f(2)=7). Since no neighbor offers a higher value, the algorithm terminates at this local maximum. This algorithm reaches a local maximum at (x=2) but misses the global maximum at (x=4) ((f(4)=9)), which is common in hill-climbing due to the discrete landscape and neighbor definition	
4 b	Genetic Algorithm (GA) is a heuristic search algorithm inspired by the process of natural selection. It is used to find optimal or near-optimal solutions to difficult problems. The process typically involves an initial population evolving through repeated application of specific genetic operators. **The Process of a Genetic Algorithm** The overall process follows these steps: **Initialization**: An initial population of individuals (potential solutions) is randomly generated. Each individual is usually represented as a fixed-length string, often binary. **Fitness Evaluation**: The fitness of each individual in the current population is evaluated using a predefined fitness function that measures how "good" a potential solution. **Selection**: Individuals are selected from the current population based on their fitness. Fitter individuals have a higher probability of being chosen as "parents" for the next generation. **Reproduction (Crossover and Mutation)**: Genetic operators (crossover and mutation) are applied to the selected parents to create a new generation of offspring. This introduces variation and explores new areas of the search space. **Replacement**: The new generation replaces all or some of the old population. **Termination**: The process repeats (steps 2-5) until a termination condition is met, such as a maximum number of generations, a satisfactory solution is found, or the population diversity is too low	5
5 a	Simulated Annealing (SA) is a metaheuristic optimization algorithm inspired by the physical process of annealing in metallurgy. Its core principle is to explore the search space for a global optimum by allowing occasional moves to inferior solutions, thereby avoiding entrapment in local optima. The algorithm operates on an analogy where the optimization problem corresponds to finding a low-energy state of a physical system, the objective function is analogous to energy, and the algorithm's control parameter, "temperature," governs the exploration-exploitation balance.	5

	The mechanism involves iteratively evaluating neighboring solutions. The key feature is the acceptance criterion: a superior solution (one that improves the objective function) is always accepted, while an inferior one is accepted with a certain probability that decreases as the temperature (which is gradually lowered according to a "cooling schedule") decreases. This probabilistic acceptance allows the algorithm to escape local optima. SA guarantees eventual convergence to the global optimum if the temperature decreases slowly enough (a condition known as the logarithmic cooling schedule). This slow cooling schedule ensures that the system has sufficient time at each temperature level to explore the search space thoroughly, statistically allowing it to find the globally optimal solution with a probability approaching 1.		
5 b	Local Search Algorithms are optimization techniques that iteratively improve a single candidate solution by exploring its "neighborhood" (slightly modified versions) to find better states, rather than searching the entire vast space, making them crucial for large problems where exhaustive searches are impossible. They are essential for large search spaces because they focus computational effort on promising regions, using heuristics to quickly converge to near-optimal solutions for complex real-world problems like scheduling, routing, and resource allocation, where finding the absolute best solution is computationally infeasible		5
	Definition: A local search algorithm starts with an initial solution and repeatedly makes small, incremental changes (moves to neighbors) to find progressively better solutions, guided by an objective or cost function, continuing until no further improvement can be found or a stopping condition is met. **Essential for Large Spaces:** In huge search spaces (e.g., traveling salesman problem), exhaustively checking all possibilities is intractable; local search algorithms bypass this by intelligently navigating, finding good-enough solutions efficiently by exploiting local information, making them practical for AI applications in scheduling, routing, and machine learning		